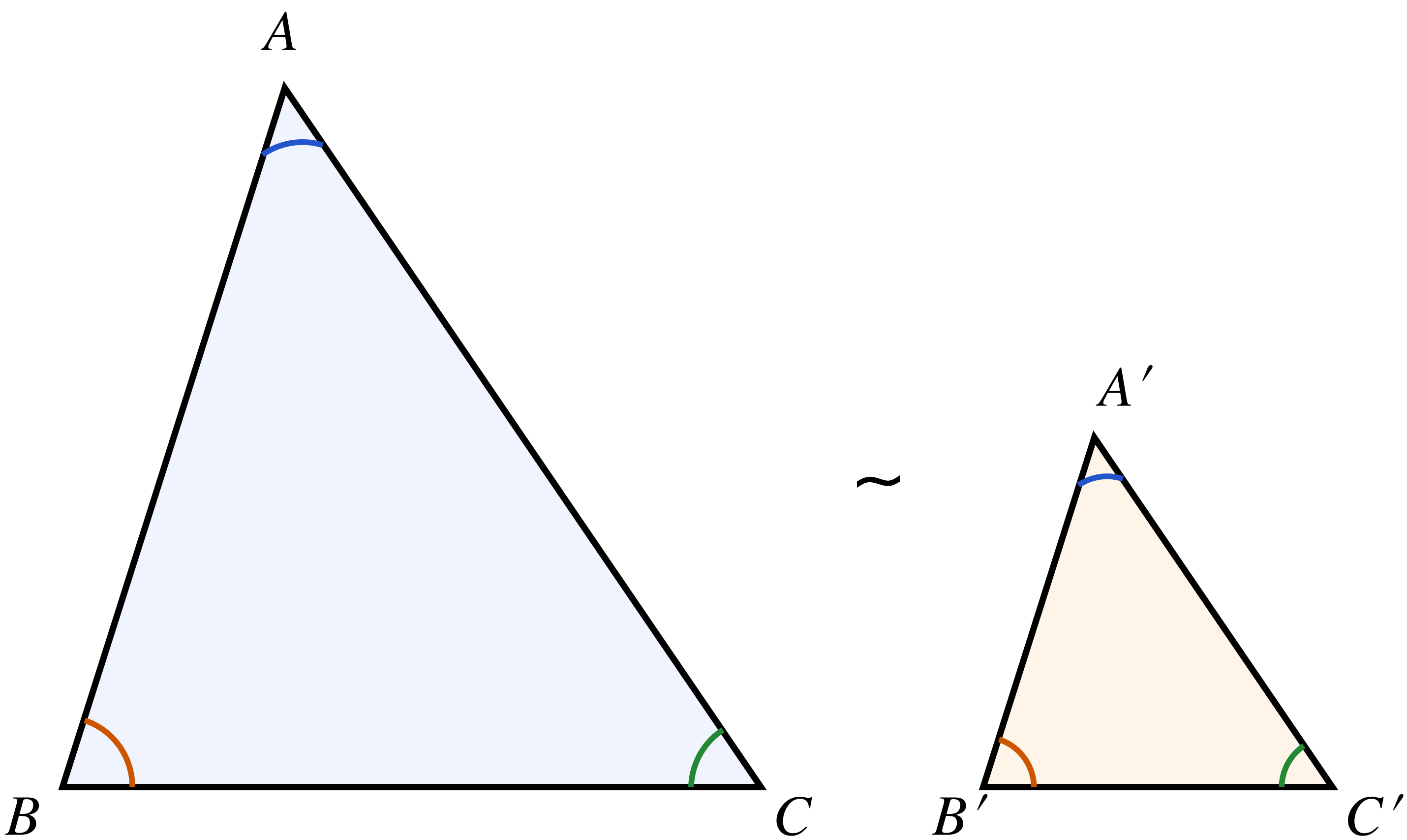
Lecture 14: Similar Triangles

Definition

Two triangles ABC and $A'B'C'$ are **similar**, written $\triangle ABC \sim \triangle A'B'C'$, if their corresponding angles are equal:

$$\angle A = \angle A', \quad \angle B = \angle B', \quad \angle C = \angle C'.$$



Theorem 1 (AA): Two triangles are similar if two pairs of corresponding angles are equal.

Proof: The angle sum in any triangle is 180°, so equality of two pairs forces the third pair to be equal as well. ■

Theorem 2: If $\triangle ABC \sim \triangle A'B'C'$, then the corresponding sides are proportional:

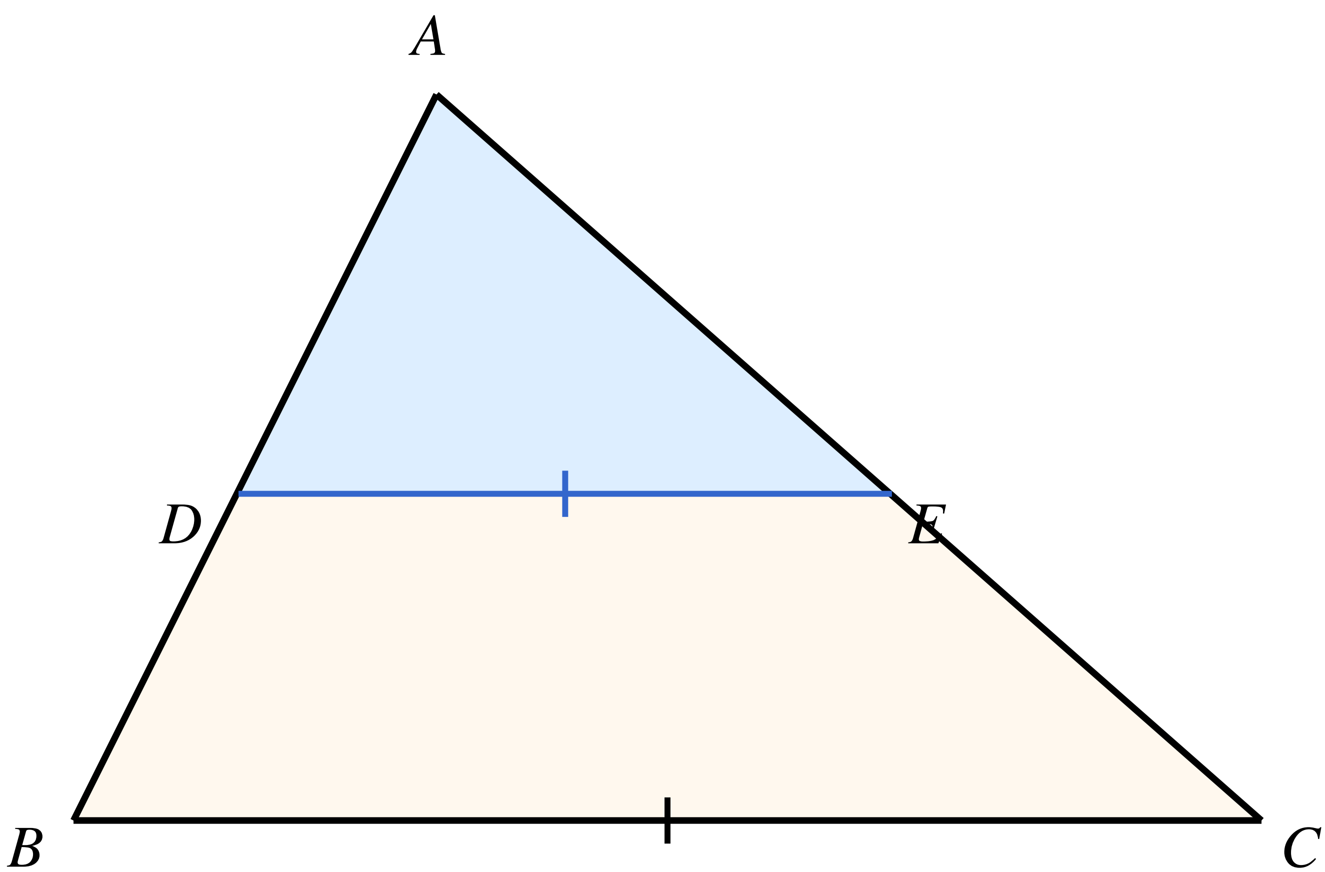
$$\frac{AB}{A'B'} = \frac{BC}{B'C'} = \frac{CA}{C'A'}.$$

The common ratio is called the **scale factor** k.

This theorem is proved using the following lemma by moving one triangle on top of the other and applying the Basic Proportionality Theorem (also known as Thales' theorem).

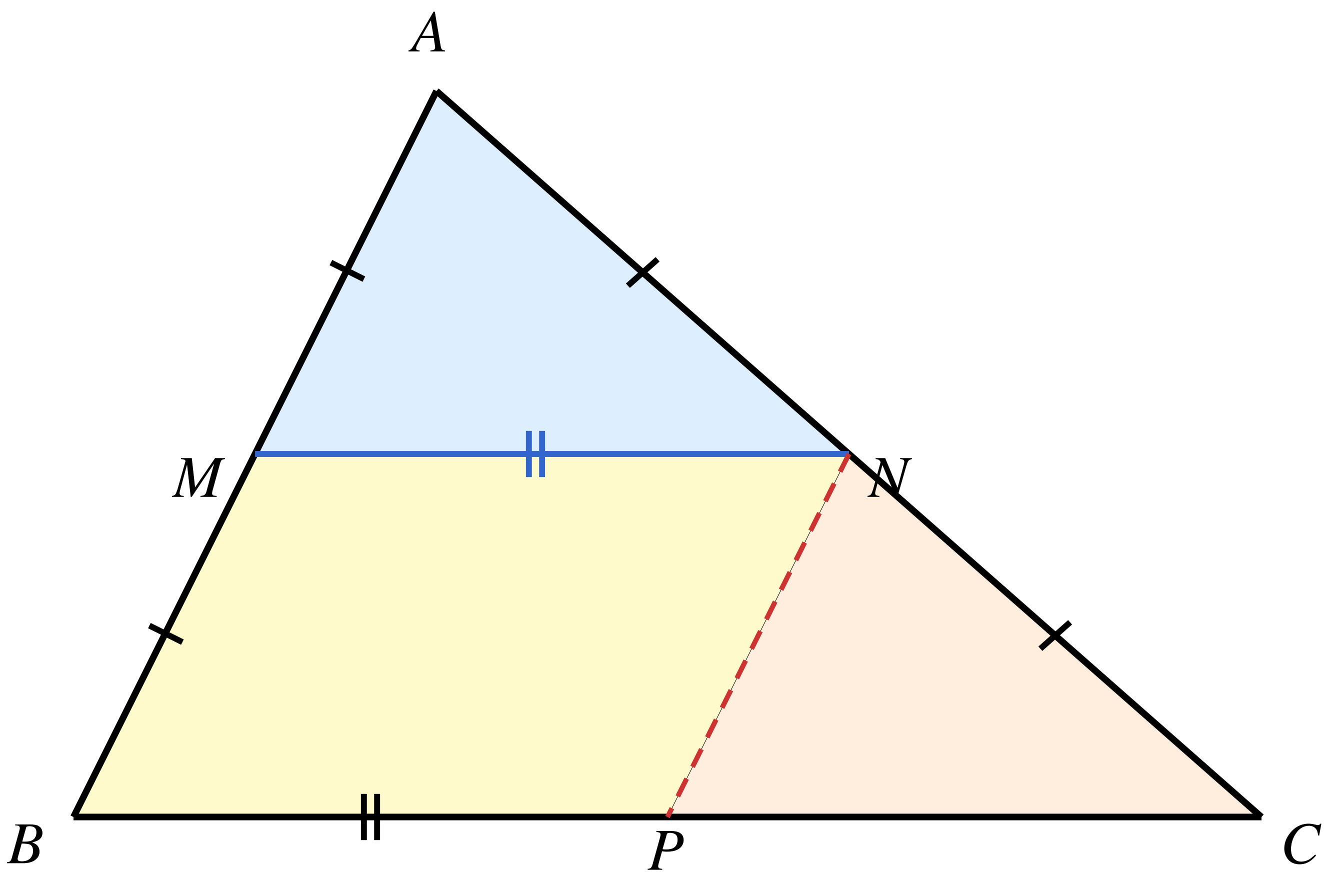
Lemma (Basic Proportionality Theorem / Thales): Let $\triangle ABC$ be a triangle. If a line parallel to BC meets AB at D and AC at E, then

$$\frac{AD}{AB} = \frac{AE}{AC} = \frac{DE}{BC}.$$



We only prove the basic proportionality theorem in a special case.

Special case (Bimedian Theorem): Let M be the midpoints of side AB of $\triangle ABC$. If $MN \parallel BC$, then $MN = \dfrac{1}{2}BC$ and N is the mid point of AC.



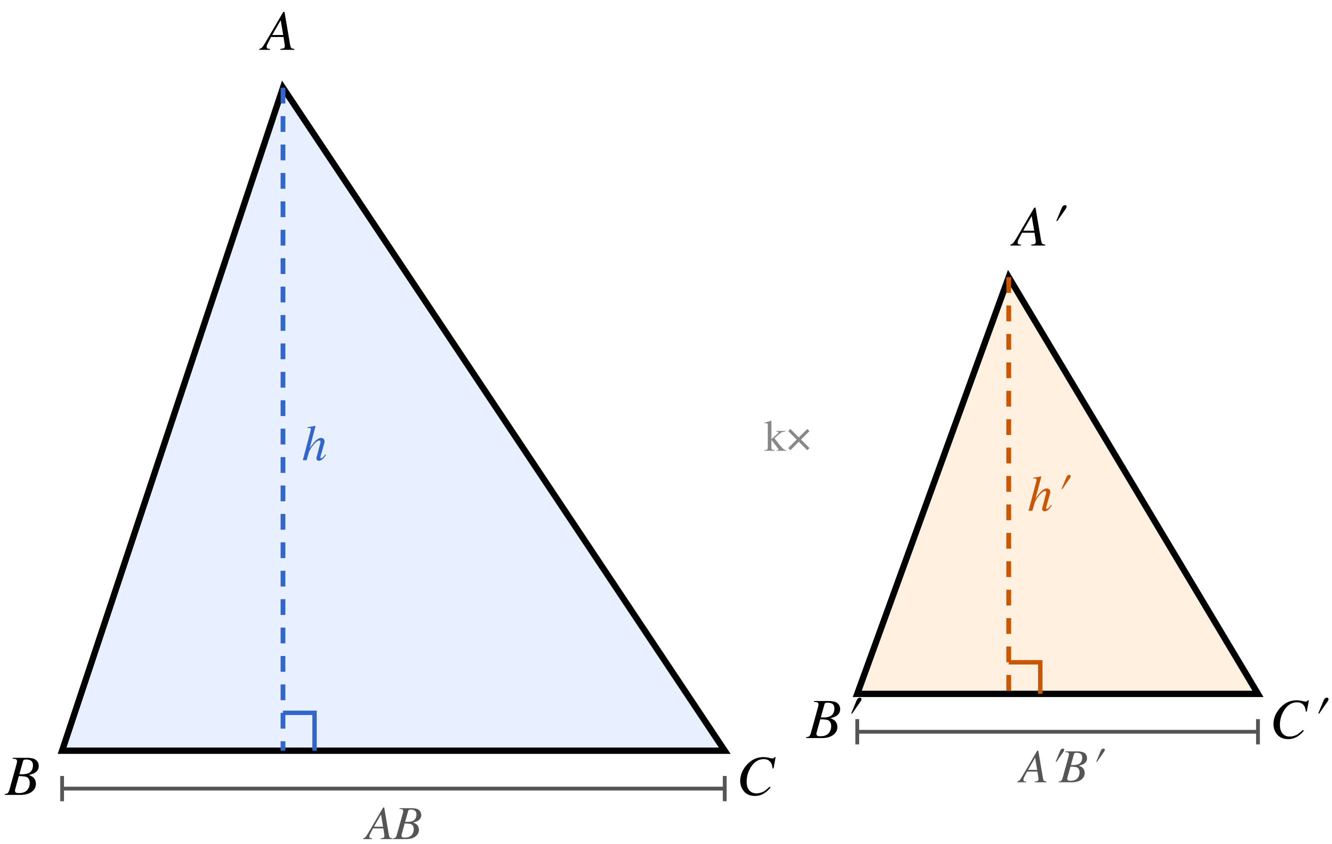
Proof:

- Construct the parallel line to AB through N. So we get a parallelogram $BPNM$. It implies that $MN = BP$ and $BM = NP$.
- Since $NP \parallel AB$, we have $\angle CAB = \angle CNP$ and $\angle ABC = \angle NPC$. By AAS, $\triangle AMN \cong \triangle NPC$. In particular, $AN = NP$ and $PC = MN$.
- This implies that N is the midpoint of AC and $MN = BP = PC = \frac{1}{2}BC$. ■

Theorem 3: Area Ratio

Theorem 3: If $\triangle ABC \sim \triangle A'B'C'$ with scale factor k, then

$$\frac{S_{\triangle ABC}}{S_{\triangle A'B'C'}} = k^2.$$



Proof:

- Let h and h' be the altitudes from C and C' to AB and $A'B'$ respectively. The foot of each altitude together with the two base vertices forms a right triangle. Since $\angle A = \angle A'$, these right triangles are similar with ratio k, giving $h/h' = k$.
- Therefore:

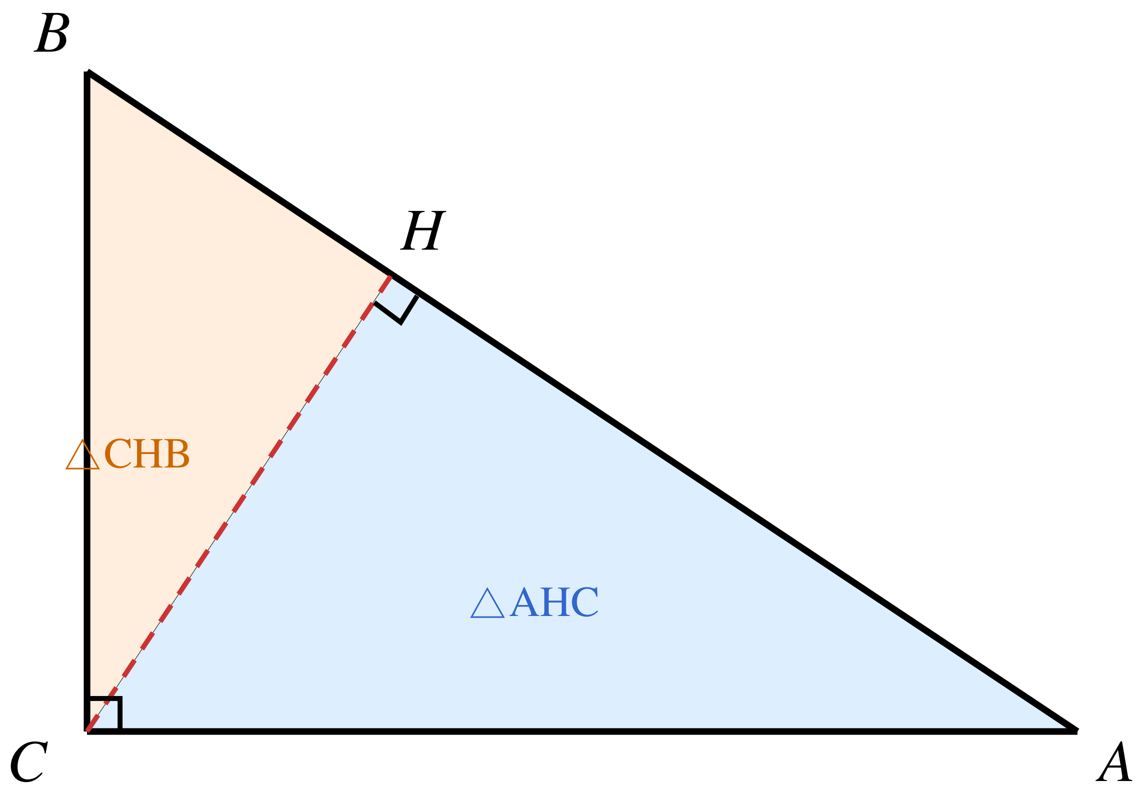
$$\frac{S_{\triangle ABC}}{S_{\triangle A'B'C'}} = \frac{\frac{1}{2} \cdot AB \cdot h}{\frac{1}{2} \cdot A'B' \cdot h'} = \frac{AB}{A'B'} \cdot \frac{h}{h'} = k \cdot k = k^2.$$

■

Theorem 4: Right Triangle Altitude Theorem

Theorem 4: In a right triangle ABC with $\angle C = 90^\circ$, let H be the foot of the altitude from C to the hypotenuse AB. Then:

1. $\triangle ACB \sim \triangle AHC \sim \triangle CHB$.
2. $CH^2 = AH \cdot HB$.
3. $AC^2 = AH \cdot AB$ and $BC^2 = BH \cdot AB$.



Proof:

- In $\triangle AHC$ and $\triangle ACB$: $\angle A$ is shared and $\angle AHC = \angle ACB = 90^\circ$. By AA, $\triangle AHC \sim \triangle ACB$.
- In $\triangle CHB$ and $\triangle ACB$: $\angle B$ is shared and $\angle CHB = 90^\circ$. By AA, $\triangle CHB \sim \triangle ACB$.
- Hence all three triangles are mutually similar. Reading off the proportional sides:

$$\frac{AH}{AC} = \frac{AC}{AB} \implies AC^2 = AH \cdot AB,$$

$$\frac{BH}{BC} = \frac{BC}{AB} \implies BC^2 = BH \cdot AB,$$

$$\frac{AH}{CH} = \frac{CH}{HB} \implies CH^2 = AH \cdot HB.$$

■

Corollary 1: The altitude CH is the geometric mean of the two segments AH and HB into which it divides the hypotenuse.

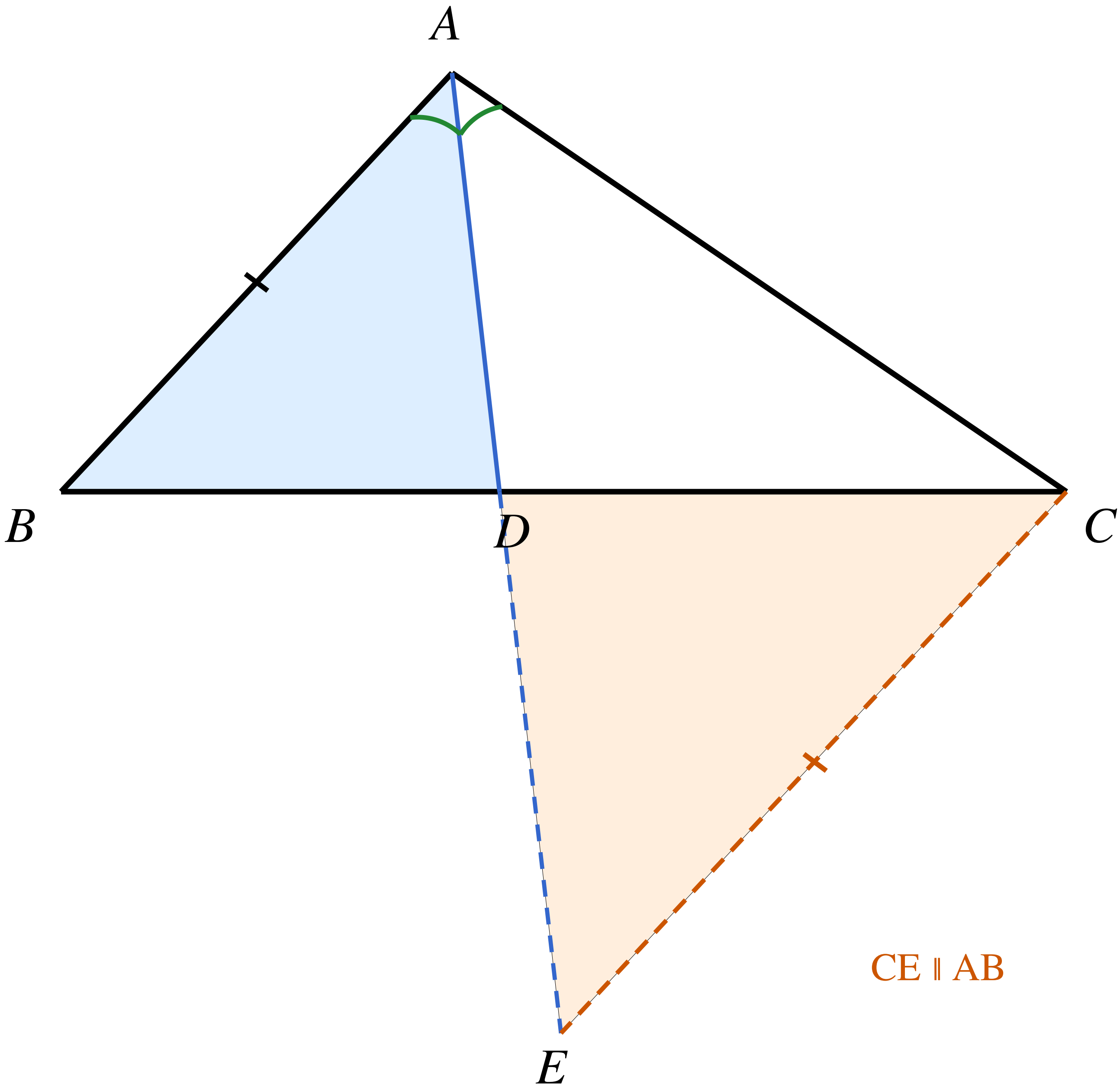
Pythagorean Theorem:

$$AC^2 + BC^2 = AB^2.$$

Proof: From the above, we have $AC^2 + BC^2 = AH \cdot AB + BH \cdot AB = AB \cdot (AH + BH) = AB^2$.

Bisector Theorem: In $\triangle ABC$, if AD is the bisector of $\angle A$ meeting BC at D, then

$$\frac{BD}{DC} = \frac{AB}{AC}.$$



Proof: Construct the parallel line to AB through C. Let it meet the extension of AD at E. Then $\angle BAC = \angle ECD$ and $\angle ABC = \angle EDC$. By AA, we have

$$\triangle ABD \sim \triangle ECD$$

giving

$$\frac{BD}{DC} = \frac{AB}{EC}.$$

Now the bisection of $\angle A$ implies that $\angle BAD = \angle DAC$. So $\angle DAC = \angle AEC$. The isoceles triangle AEC has $AC = CE$. Therefore

$$\frac{BD}{DC} = \frac{AB}{AC}.$$